APT THREAT HUNTING IN RAIL TRANSPORTATION

MITRE ATT&CK | Threat Intelligence | Detection Engineering

Prepared By

Martins Adeyanju (Cybersecurity Analyst)

PROJECT OVERVIEW

This project presents a structured threat hunting investigation focused on APT groups targeting the UK rail transportation sector.

INTRODUCTION

Rail infrastructure combines IT and OT systems, making it a high-value target for cyber threats.

AIMS AND APPROACH

Identify APTs, map TTPs using MITRE ATT&CK, and develop detection strategies.

SCOPE OF ANALYSIS

Country: United Kingdom
Sector: Rail Transportation
Focus: Disruption, Espionage, Credential Theft

OBJECTIVES

- Identify threat actors
- Map TTPs
- Analyze malware and CVEs
- Develop hypotheses

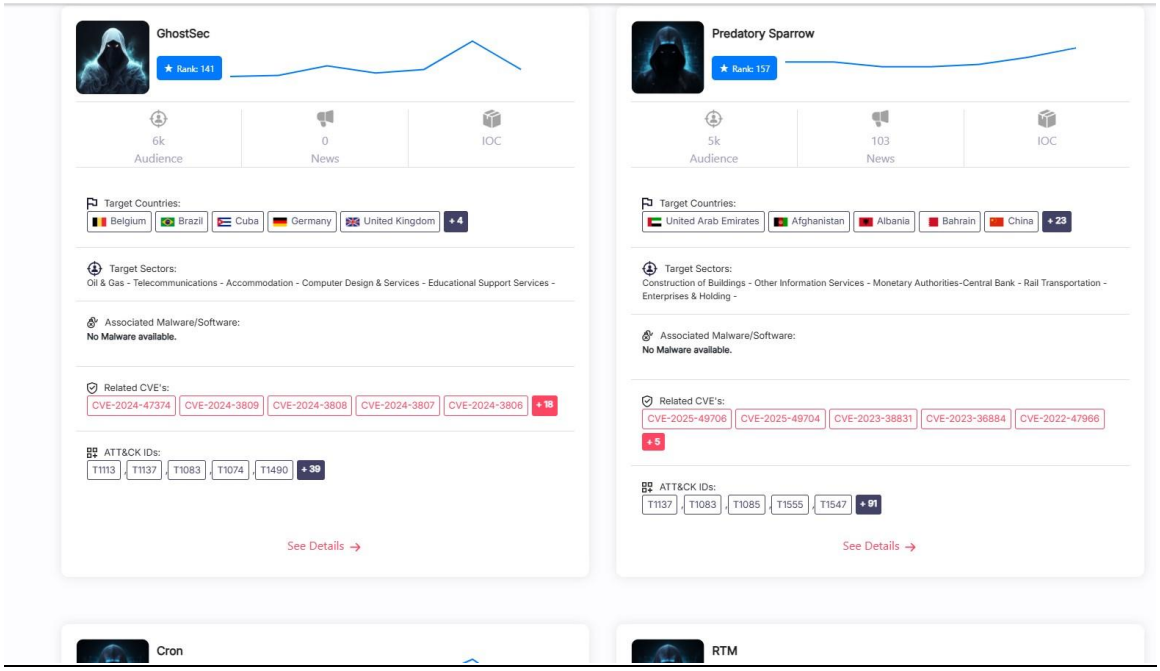
TOOLS AND RESOURCES

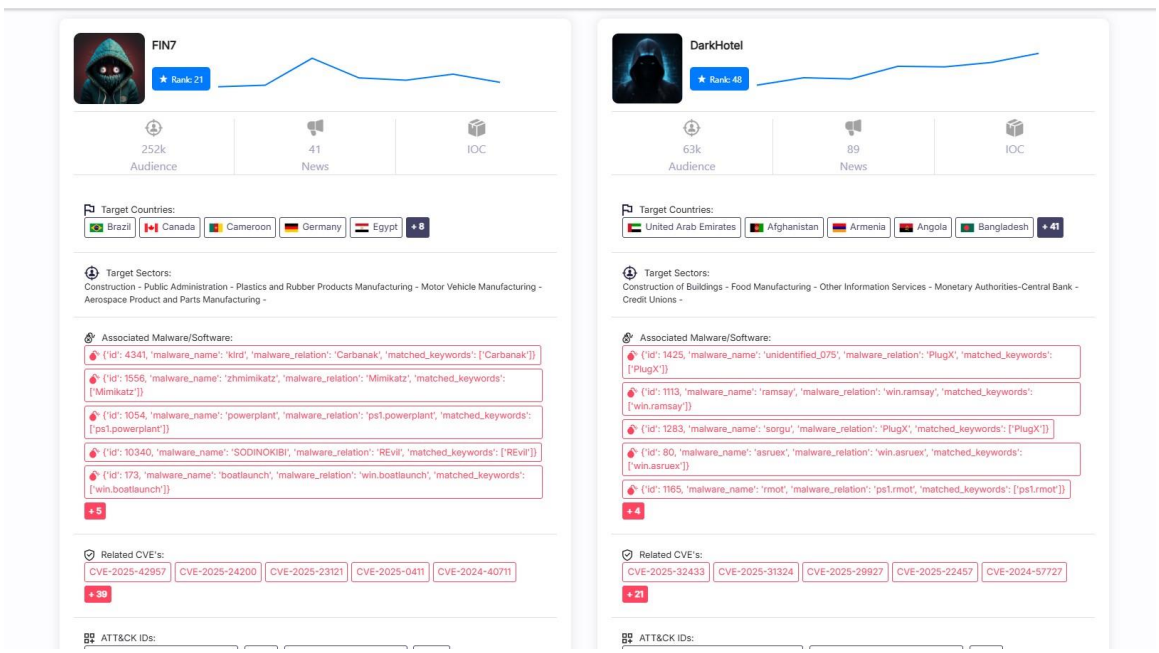
MITRE ATT&CK Navigator, SOCRadar, OSINT

APT ACTORS IDENTIFIED

FIN7, Electrum, DarkHotel, GhostSec, SideCopy, Predatory Sparrow

THREAT INTELLIGENCE DASHBOARD





ACTOR ANALYSIS

FIN7

★ Rank: 21

252k	41	IOC
Audience	News	

Target Countries:

Brazil | Canada | Cameroon | Germany | Egypt | + 8

Target Sectors:

Construction - Public Administration - Plastics and Rubber Products Manufacturing - Motor Vehicle Manufacturing - Aerospace Product and Parts Manufacturing -

Associated Malware/Software:

{'id': 4341, 'malware_name': 'klrd', 'malware_relation': 'Carbanak', 'matched_keywords': ['Carbanak']}

{'id': 1556, 'malware_name': 'zhmimikatz', 'malware_relation': 'Mimikatz', 'matched_keywords': ['Mimikatz']}

{'id': 1054, 'malware_name': 'powerplant', 'malware_relation': 'ps1.powerplant', 'matched_keywords': ['ps1.powerplant']}

{'id': 10340, 'malware_name': 'SODINOKIBI', 'malware_relation': 'REvil', 'matched_keywords': ['REvil']}

{'id': 173, 'malware_name': 'boatlaunch', 'malware_relation': 'win.boatlaunch', 'matched_keywords': ['win.boatlaunch']}

+ 5

Related CVE's:

CVE-2025-42957 | CVE-2025-24200 | CVE-2025-23121 | CVE-2025-0411 | CVE-2024-40711

+ 39

ATT&CK IDs:

DarkHotel

★ Rank: 48

63k	89	IOC
Audience	News	

Target Countries:

United Arab Emirates | Afghanistan | Armenia | Angola | Bangladesh | + 41

Target Sectors:

Construction of Buildings - Food Manufacturing - Other Information Services - Monetary Authorities-Central Bank - Credit Unions -

Associated Malware/Software:

{'id': 1425, 'malware_name': 'unidentified_075', 'malware_relation': 'PlugX', 'matched_keywords': ['PlugX']}

{'id': 1113, 'malware_name': 'ramsay', 'malware_relation': 'win.ramsay', 'matched_keywords': ['win.ramsay']}

{'id': 1283, 'malware_name': 'sorgu', 'malware_relation': 'PlugX', 'matched_keywords': ['PlugX']}

{'id': 80, 'malware_name': 'asruex', 'malware_relation': 'win.asruex', 'matched_keywords': ['win.asruex']}

{'id': 1165, 'malware_name': 'rmot', 'malware_relation': 'ps1.rmot', 'matched_keywords': ['ps1.rmot']}

+ 4

Related CVE's:

CVE-2025-32433 | CVE-2025-31324 | CVE-2025-29927 | CVE-2025-22457 | CVE-2024-57727

+ 21

GhostSec

★ Rank: 141

6k	0	IOC
Audience	News	

Target Countries:

Belgium | Brazil | Cuba | Germany | United Kingdom | + 4

Target Sectors:

Oil & Gas - Telecommunications - Accommodation - Computer Design & Services - Educational Support Services -

Associated Malware/Software:

No Malware available.

Related CVE's:

CVE-2024-47374 | CVE-2024-3809 | CVE-2024-3808 | CVE-2024-3807 | CVE-2024-3806 | + 18

ATT&CK IDs:

T1113 , T1137 , T1083 , T1074 , T1490 | + 39

See Details →

Predatory Sparrow

★ Rank: 157

5k	103	IOC
Audience	News	

Target Countries:

United Arab Emirates | Afghanistan | Albania | Bahrain | China | + 23

Target Sectors:

Construction of Buildings - Other Information Services - Monetary Authorities-Central Bank - Rail Transportation - Enterprises & Holding -

Associated Malware/Software:

No Malware available.

Related CVE's:

CVE-2025-49706 | CVE-2025-49704 | CVE-2023-38831 | CVE-2023-36884 | CVE-2022-47966 | + 5

ATT&CK IDs:

T1137 , T1083 , T1085 , T1555 , T1547 | + 91

See Details →

ELECTRUM

★ Rank: 39

91k Audience	82 News	IOC

Target Countries:

United Arab Emirates | Angola | Argentina | Australia | Bangladesh | + 24

Target Sectors:

Agriculture&Forestry - Energy & Utilities - Retail - Transportation&Warehousing - Finance -

Associated Malware/Software:

{'id': 580, 'malware_name': 'grey_energy', 'malware_relation': 'GreyEnergy', 'matched_keywords': ['GreyEnergy', 'win.grey_energy']}

{'id': 613, 'malware_name': 'hermeticwiper', 'malware_relation': 'KillDisk', 'matched_keywords': ['KillDisk', 'win.killdisk']}

{'id': 72, 'malware_name': 'arguepatch', 'malware_relation': 'win.arguepatch', 'matched_keywords': ['win.arguepatch']}

{'id': 1362, 'malware_name': 'teledoor', 'malware_relation': 'win.teledoor', 'matched_keywords': ['win.teledoor']}

{'id': 459, 'malware_name': 'exaramel', 'malware_relation': 'elf.exaramel', 'matched_keywords': ['elf.exaramel', 'win.exaramel', 'Exaramel for Linux', 'Exaramel for Windows']}

+ 10

Related CVE's:

CVE-2026-24858 | CVE-2025-8088 | CVE-2025-64328 | CVE-2025-59719 | CVE-2025-59718

+ 42

ATT&CK IDs:

SideCopy

★ Rank: 147

5k	42	IOC
Audience	News	

Target Countries:

Hungary | India | United States

Target Sectors:

National Security and International Affairs - Data Processing, Hosting, and Related Services - Computer Systems Design and Related Services - Advertising Agencies - Computer Systems Design Services -

Associated Malware/Software:

SideCopy | win.curlback | elf.unidentified_005 | win.drat | win.action_rat | + 2

Related CVE's:

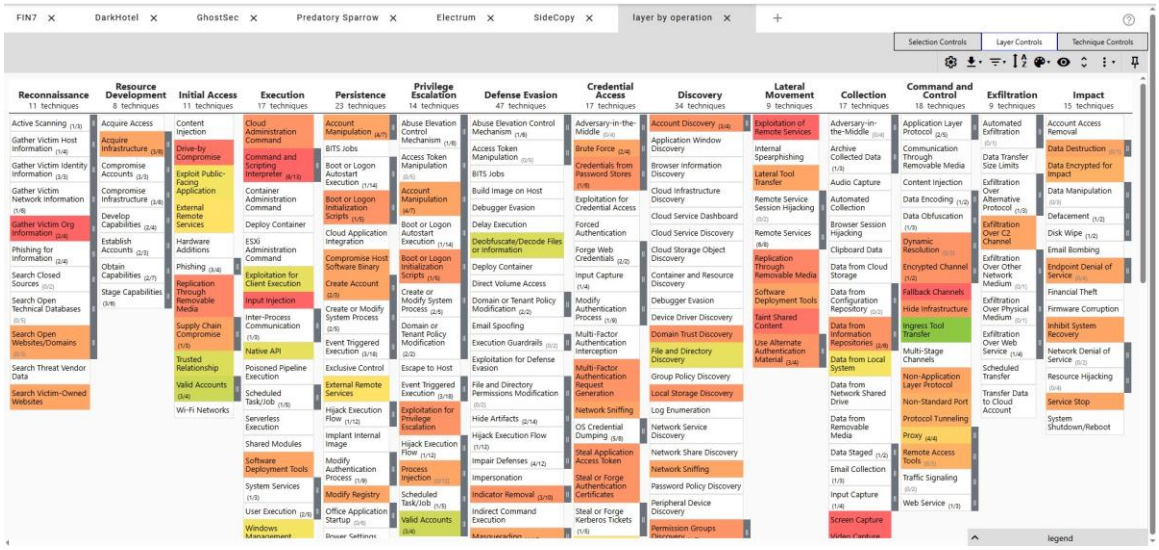
CVE-2024-3094 | CVE-2024-21893 | CVE-2024-21887 | CVE-2023-46805 | CVE-2023-38831 | + 7

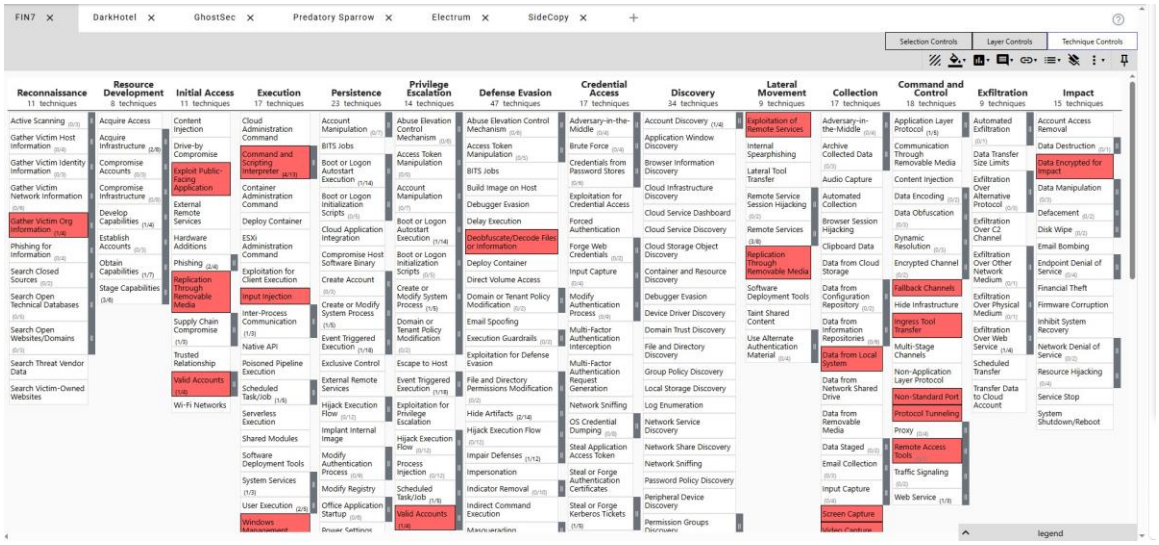
ATT&CK IDs:

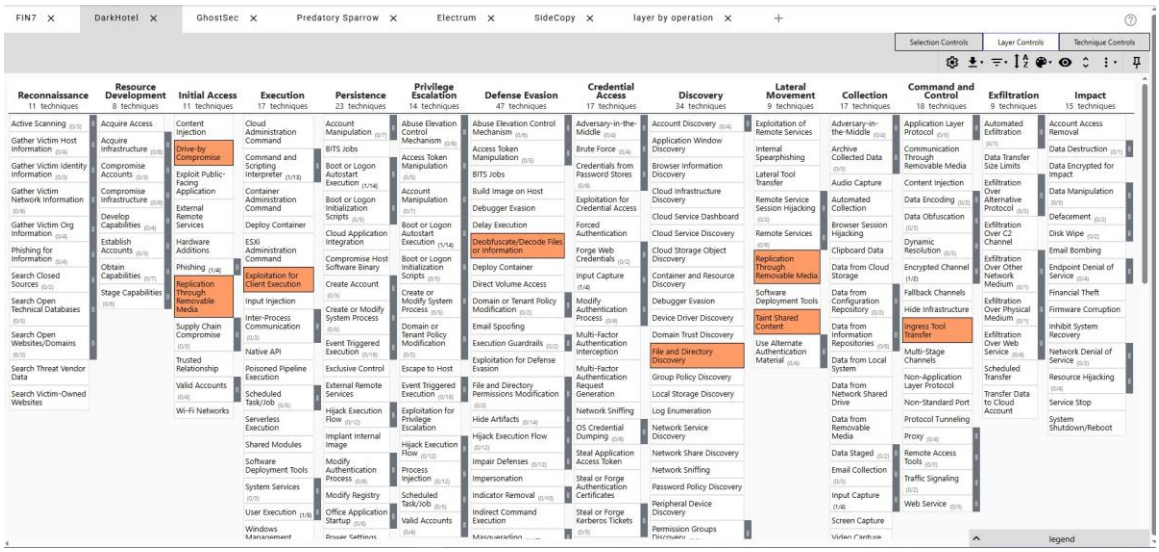
T1189 , T1170 , T1082 - System Information Discovery , T1056 , T1574.001 | + 102

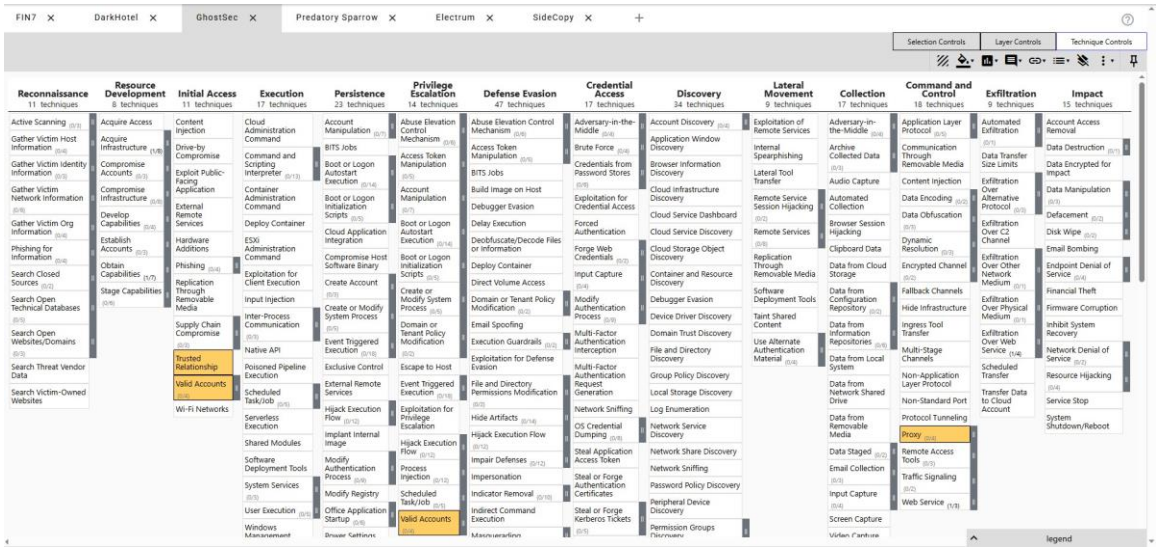
See Details →

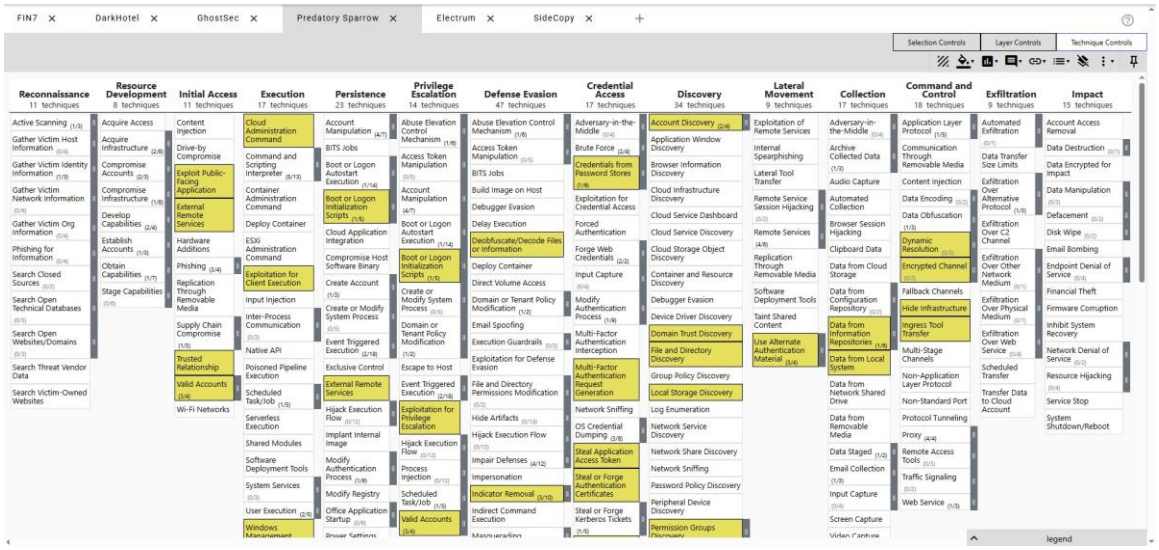
MITRE ATT&CK NAVIGATOR

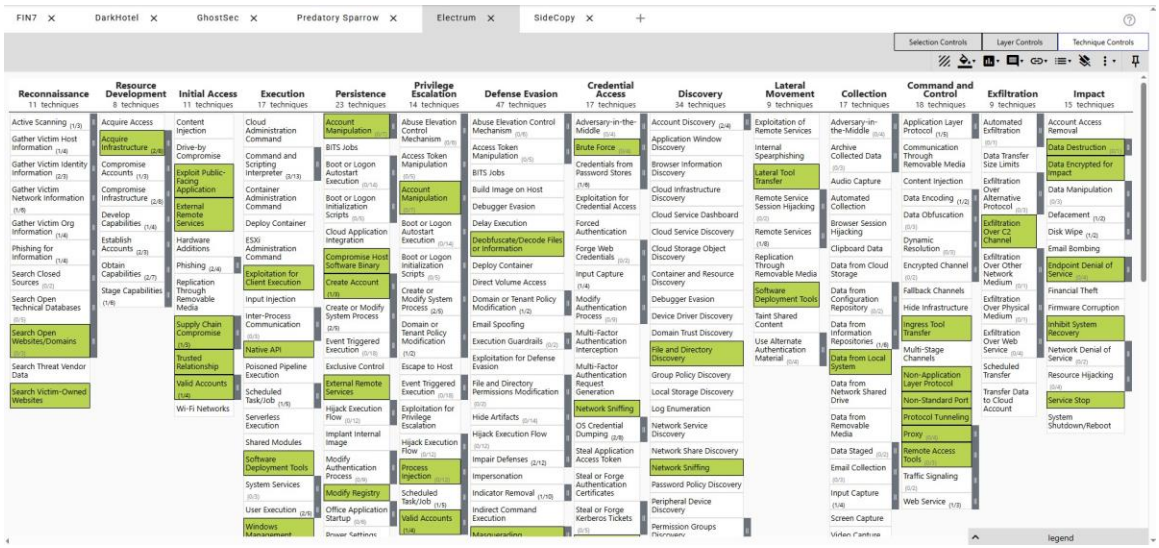


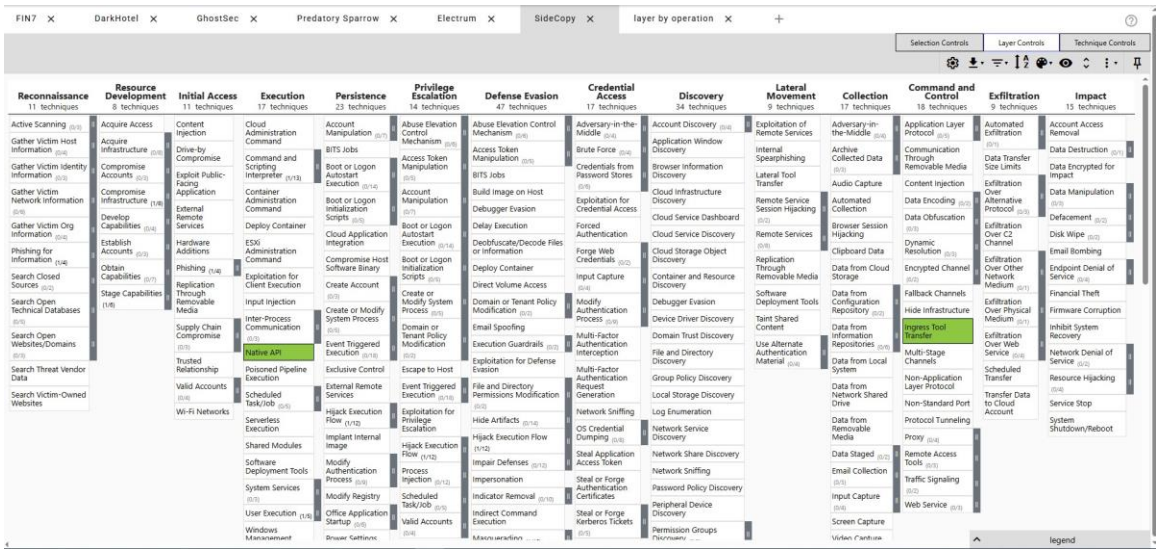



FIN7
DarkHotel
GhostSec
Predatory Sparrow
Electrum
SideCopy
layer by operation
Selection Controls
Layer Controls
Technique Controls
Reconnaissance 11 techniques
Active Scanning (0/3)
Gather Victim Host Information (0/4)
Gather Victim Identity Information (0/3)
Gather Victim Network Information (0/6)
Gather Victim Org Information (0/4)
Phishing for Information (0/4)
Search Closed Sources (0/2)
Search Open Technical Databases (0/5)
Search Open Websites/Domains (0/3)
Search Threat Vendor Data
Search Victim-Owned Websites
Resource Development 8 techniques
Acquire Access
Acquire Infrastructure (0/8)
Compromise Accounts (0/3)
Compromise Infrastructure (0/6)
Develop Capabilities (0/4)
Establish Accounts (0/3)
Obtain Capabilities (0/7)
Stage Capabilities (0/6)
Initial Access 11 techniques
Content Injection
Drive-by Compromise
Exploit Public-Facing Application
External Remote Services
Hardware Additions
Phishing (1/4)
Replication Through Removable Media
Supply Chain Compromise (0/3)
Trusted Relationship
Valid Accounts (0/4)
Wi-Fi Networks
Execution 17 techniques
Cloud Administration Command
Command and Scripting Interpreter (1/13)
Container Administration Command
Deploy Container
ESXi Administration Command
Exploitation for Client Execution
Input Injection
Inter-Process Communication (0/3)
Native API
Poisoned Pipeline Execution
Scheduled Task/Job (0/5)
Serverless Execution
Shared Modules
Software Deployment Tools
System Services (0/3)
User Execution (1/3)
Windows Management
Persistence 23 techniques
Account Manipulation (0/7)
BITS Jobs
Boot or Logon Autostart Execution (1/14)
Boot or Logon Initialization Scripts (0/5)
Cloud Application Integration
Compromise Host Software Binary
Create Account (0/3)
Create or Modify System Process (0/5)
Event Triggered Execution (0/16)
Exclusive Control
External Remote Services
Hijack Execution Flow (0/12)
Implant Internal Image
Modify Authentication Process (0/8)
Modify Registry
Office Application Startup (0/6)
Browser Settings
Privilege Escalation 14 techniques
Abuse Elevation Control Mechanism (0/6)
Access Token Manipulation (0/5)
Account Manipulation (0/5)
Boot or Logon Autostart Execution (1/14)
Boot or Logon Initialization Scripts (0/5)
Create or Modify System Process (0/5)
Domain or Tenant Policy Modification (0/2)
Escape to Host
Event Triggered Execution (0/16)
Exploitation for Privilege Escalation
Hijack Execution Flow (0/12)
Process Injection (0/12)
Scheduled Task/Job (0/5)
Valid Accounts (0/4)
Defense Evasion 47 techniques
Abuse Elevation Control Mechanism (0/6)
Access Token Manipulation (0/5)
BITS Jobs
Build Image on Host
Debugger Evasion
Delay Execution
Deobfuscate/Decode Files or Information
Deploy Container
Direct Volume Access
Domain or Tenant Policy Modification (0/2)
Email Spoofing
Execution Guardrails (0/2)
Exploitation for Defense Evasion
File and Directory Permissions Modification (0/2)
Hide Artifacts (0/14)
Hijack Execution Flow (0/12)
Impair Defenses (0/12)
Impersonation
Indicator Removal (0/10)
Indirect Command Execution
Masquerading
Credential Access 17 techniques
Adversary-in-the-Middle (0/4)
Brute Force (0/4)
Credentials from Password Stores (0/6)
Exploitation for Credential Access
Forced Authentication
Forge Web Credentials (0/2)
Input Capture (1/4)
Modify Authentication Process (0/9)
Multi-Factor Authentication Interception
Multi-Factor Authentication Request Generation
Network Sniffing
OS Credential Dumping (0/8)
Steal Application Access Token
Steal or Forge Authentication Certificates
Steal or Forge Kerberos Tickets (0/5)
Discovery 34 techniques
Account Discovery (0/4)
Application Window Discovery
Browser Information Discovery
Cloud Infrastructure Discovery
Cloud Service Dashboard
Cloud Service Discovery
Cloud Storage Object Discovery
Container and Resource Discovery
Debugger Evasion
Device Driver Discovery
Domain Trust Discovery
File and Directory Discovery
Group Policy Discovery
Local Storage Discovery
Log Enumeration
Network Service Discovery
Network Share Discovery
Network Sniffing
Password Policy Discovery
Peripheral Device Discovery
Permission Groups Discovery
Lateral Movement 9 techniques
Exploitation of Remote Services
Internal Spearphishing
Lateral Tool Transfer
Remote Service Session Hijacking (0/2)
Remote Services (0/6)
Replication Through Removable Media
Software Deployment Tools
Taint Shared Content
Use Alternate Authentication Material (0/4)
Collection 17 techniques
Adversary-in-the-Middle (0/4)
Archive Collected Data (0/3)
Audio Capture
Automated Collection
Browser Session Hijacking
Clipboard Data
Data from Cloud Storage
Data from Configuration Repository (0/2)
Data from Information Repositories (0/3)
Data from Local System
Data from Network Shared Drive
Data from Removable Media
Data Staged (0/2)
Email Collection (0/3)
Input Capture (1/4)
Screen Capture
Video Capture
Command and Control 18 techniques
Application Layer Protocol (0/5)
Communication Through Removable Media
Content Injection
Data Encoding (0/2)
Data Obfuscation (0/3)
Dynamic Resolution (0/3)
Encrypted Channel (1/2)
Fallback Channels
Hide Infrastructure
Ingress Tool Transfer
Multi-Stage Channels
Non-Application Layer Protocol
Non-Standard Port
Protocol Tunneling
Proxy (0/4)
Remote Access Tools (0/3)
Traffic Signaling (0/2)
Web Service (0/3)
Exfiltration 9 techniques
Automated Exfiltration (0/1)
Data Transfer Size Limits
Exfiltration Over Alternative Protocol (0/3)
Exfiltration Over C2 Channel
Exfiltration Over Other Network Medium (0/1)
Exfiltration Over Physical Medium (0/1)
Exfiltration Over Web Service (0/4)
Scheduled Transfer
Transfer Data to Cloud Account
Impact 15 techniques
Account Access Removal
Data Destruction (0/1)
Data Encrypted for Impact
Data Manipulation (0/3)
Defacement (0/2)
Disk Wipe (0/2)
Email Bombing
Endpoint Denial of Service (0/4)
Financial Theft
Firmware Corruption
Inhibit System Recovery
Network Denial of Service (0/2)
Resource Hijacking (0/4)
Service Stop
System Shutdown/Reboot
legend


FIN7
DarkHotel
GhostSec
Predatory Sparrow
Electrum
SideCopy
Selection Controls
Layer Controls
Technique Controls
Reconnaissance 11 techniques
Active Scanning (0/3)
Gather Victim Host Information (0/4)
Gather Victim Identity Information (0/3)
Gather Victim Network Information (0/6)
Gather Victim Org Information (0/4)
Phishing for Information (0/4)
Search Closed Sources (0/2)
Search Open Technical Databases (0/5)
Search Open Websites/Domains (0/3)
Search Threat Vendor Data
Search Victim-Owned Websites
Resource Development 8 techniques
Acquire Access
Acquire Infrastructure (1/8)
Compromise Accounts (0/3)
Compromise Infrastructure (0/6)
Develop Capabilities (0/4)
Establish Accounts (0/3)
Obtain Capabilities (1/7)
Stage Capabilities (0/6)
Initial Access 11 techniques
Content Injection
Drive-by Compromise
Exploit Public-Facing Application
External Remote Services
Hardware Additions
Phishing (0/4)
Replication Through Removable Media
Supply Chain Compromise (0/3)
Trusted Relationship
Valid Accounts (0/4)
Wi-Fi Networks
Execution 17 techniques
Cloud Administration Command
Command and Scripting Interpreter (0/13)
Container Administration Command
Deploy Container
ESXi Administration Command
Exploitation for Client Execution
Input Injection
Inter-Process Communication (0/3)
Native API
Poisoned Pipeline Execution
Scheduled Task/Job (0/5)
Serverless Execution
Shared Modules
Software Deployment Tools
System Services (0/3)
User Execution (0/3)
Windows Management
Persistence 23 techniques
Account Manipulation (0/7)
BITS Jobs
Boot or Logon Autostart Execution (0/14)
Boot or Logon Initialization Scripts (0/5)
Cloud Application Integration
Compromise Host Software Binary
Create Account (0/3)
Create or Modify System Process (0/5)
Event Triggered Execution (0/16)
Exclusive Control
External Remote Services
Hijack Execution Flow (0/12)
Implant Internal Image
Modify Authentication Process (0/8)
Modify Registry
Office Application Startup (0/6)
Browser Settings
Privilege Escalation 14 techniques
Abuse Elevation Control Mechanism (0/6)
Access Token Manipulation (0/5)
Account Manipulation (0/5)
Boot or Logon Autostart Execution (0/14)
Boot or Logon Initialization Scripts (0/5)
Create or Modify System Process (0/5)
Domain or Tenant Policy Modification (0/2)
Escape to Host
Event Triggered Execution (0/16)
Exploitation for Privilege Escalation
Hijack Execution Flow (0/12)
Process Injection (0/12)
Scheduled Task/Job (0/5)
Valid Accounts (0/4)
Defense Evasion 47 techniques
Abuse Elevation Control Mechanism (0/6)
Access Token Manipulation (0/5)
BITS Jobs
Build Image on Host
Debugger Evasion
Delay Execution
Deobfuscate/Decode Files or Information
Deploy Container
Direct Volume Access
Domain or Tenant Policy Modification (0/2)
Email Spoofing
Execution Guardrails (0/2)
Exploitation for Defense Evasion
File and Directory Permissions Modification (0/2)
Hide Artifacts (0/14)
Hijack Execution Flow (0/12)
Impair Defenses (0/12)
Impersonation
Indicator Removal (0/10)
Indirect Command Execution
Masquerading
Credential Access 17 techniques
Adversary-in-the-Middle (0/4)
Brute Force (0/4)
Credentials from Password Stores (0/6)
Exploitation for Credential Access
Forced Authentication
Forge Web Credentials (0/2)
Input Capture (0/4)
Modify Authentication Process (0/9)
Multi-Factor Authentication Interception
Multi-Factor Authentication Request Generation
Network Sniffing
OS Credential Dumping (0/8)
Steal Application Access Token
Steal or Forge Authentication Certificates
Steal or Forge Kerberos Tickets (0/5)
Discovery 34 techniques
Account Discovery (0/4)
Application Window Discovery
Browser Information Discovery
Cloud Infrastructure Discovery
Cloud Service Dashboard
Cloud Service Discovery
Cloud Storage Object Discovery
Container and Resource Discovery
Debugger Evasion
Device Driver Discovery
Domain Trust Discovery
File and Directory Discovery
Group Policy Discovery
Local Storage Discovery
Log Enumeration
Network Service Discovery
Network Share Discovery
Network Sniffing
Password Policy Discovery
Peripheral Device Discovery
Permission Groups Discovery
Lateral Movement 9 techniques
Exploitation of Remote Services
Internal Spearphishing
Lateral Tool Transfer
Remote Service Session Hijacking (0/2)
Remote Services (0/6)
Replication Through Removable Media
Software Deployment Tools
Taint Shared Content
Use Alternate Authentication Material (0/4)
Collection 17 techniques
Adversary-in-the-Middle (0/4)
Archive Collected Data (0/3)
Audio Capture
Automated Collection
Browser Session Hijacking
Clipboard Data
Data from Cloud Storage
Data from Configuration Repository (0/2)
Data from Information Repositories (0/3)
Data from Local System
Data from Network Shared Drive
Data from Removable Media
Data Staged (0/2)
Email Collection (0/3)
Input Capture (0/4)
Screen Capture
Video Capture
Command and Control 18 techniques
Application Layer Protocol (0/5)
Communication Through Removable Media
Content Injection
Data Encoding (0/2)
Data Obfuscation (0/3)
Dynamic Resolution (0/3)
Encrypted Channel (0/2)
Fallback Channels
Hide Infrastructure
Ingress Tool Transfer
Multi-Stage Channels
Non-Application Layer Protocol
Non-Standard Port
Protocol Tunneling
Proxy (0/4)
Remote Access Tools (0/3)
Traffic Signaling (0/2)
Web Service (1/3)
Exfiltration 9 techniques
Automated Exfiltration (0/1)
Data Transfer Size Limits
Exfiltration Over Alternative Protocol (0/3)
Exfiltration Over C2 Channel
Exfiltration Over Other Network Medium (0/1)
Exfiltration Over Physical Medium (0/1)
Exfiltration Over Web Service (1/4)
Scheduled Transfer
Transfer Data to Cloud Account
Impact 15 techniques
Account Access Removal
Data Destruction (0/1)
Data Encrypted for Impact
Data Manipulation (0/3)
Defacement (0/2)
Disk Wipe (0/2)
Email Bombing
Endpoint Denial of Service (0/4)
Financial Theft
Firmware Corruption
Inhibit System Recovery
Network Denial of Service (0/2)
Resource Hijacking (0/4)
Service Stop
System Shutdown/Reboot
legend


FIN7
DarkHotel
GhostSec
Predatory Sparrow
Electrum
SideCopy
Selection Controls
Layer Controls
Technique Controls
Reconnaissance 11 techniques
Active Scanning (1/3)
Gather Victim Host Information (0/4)
Gather Victim Identity Information (1/3)
Gather Victim Network Information (0/6)
Gather Victim Org Information (0/4)
Phishing for Information (0/4)
Search Closed Sources (0/2)
Search Open Technical Databases (0/5)
Search Open Websites/Domains (0/3)
Search Threat Vendor Data
Search Victim-Owned Websites
Resource Development 8 techniques
Acquire Access
Acquire Infrastructure (2/8)
Compromise Accounts (2/3)
Compromise Infrastructure (1/6)
Develop Capabilities (2/4)
Establish Accounts (1/3)
Obtain Capabilities (1/7)
Stage Capabilities (0/6)
Initial Access 11 techniques
Content Injection
Drive-by Compromise
Exploit Public-Facing Application
External Remote Services
Hardware Additions
Phishing (3/4)
Replication Through Removable Media
Supply Chain Compromise (1/3)
Trusted Relationship
Valid Accounts (1/4)
Wi-Fi Networks
Execution 17 techniques
Cloud Administration Command
Command and Scripting Interpreter (5/13)
Container Administration Command
Deploy Container
ESXi Administration Command
Exploitation for Client Execution
Input Injection
Inter-Process Communication (0/3)
Native API
Poisoned Pipeline Execution
Scheduled Task/Job (1/5)
Serverless Execution
Shared Modules
Software Deployment Tools
System Services (0/3)
User Execution (2/3)
Windows Management
Persistence 23 techniques
Account Manipulation (4/7)
BITS Jobs
Boot or Logon Autostart Execution (1/14)
Boot or Logon Initialization Scripts (1/5)
Cloud Application Integration
Compromise Host Software Binary
Create Account (1/3)
Create or Modify System Process (0/5)
Event Triggered Execution (2/16)
Exclusive Control
External Remote Services
Hijack Execution Flow (0/12)
Implant Internal Image
Modify Authentication Process (1/9)
Modify Registry
Office Application Startup (0/6)
Power Settings
Privilege Escalation 14 techniques
Abuse Elevation Control Mechanism (1/6)
Access Token Manipulation (0/5)
Account Manipulation (4/7)
Boot or Logon Autostart Execution (1/14)
Boot or Logon Initialization Scripts (1/5)
Create or Modify System Process (0/5)
Domain or Tenant Policy Modification (1/2)
Escape to Host
Event Triggered Execution (2/16)
Exploitation for Privilege Escalation
Hijack Execution Flow (0/12)
Process Injection (0/12)
Scheduled Task/Job (1/5)
Valid Accounts (1/4)
Defense Evasion 47 techniques
Abuse Elevation Control Mechanism (1/6)
Access Token Manipulation (0/5)
BITS Jobs
Build Image on Host
Debugger Evasion
Delay Execution
Deobfuscate/Decode Files or Information
Deploy Container
Direct Volume Access
Domain or Tenant Policy Modification (1/2)
Email Spoofing
Execution Guardrails (0/2)
Exploitation for Defense Evasion
File and Directory Permissions Modification (0/2)
Hide Artifacts (0/14)
Hijack Execution Flow (0/12)
Impair Defenses (4/12)
Impersonation
Indicator Removal (3/10)
Indirect Command Execution
Masquerading
Credential Access 17 techniques
Adversary-in-the-Middle (0/4)
Brute Force (2/4)
Credentials from Password Stores (1/6)
Exploitation for Credential Access
Forced Authentication
Forge Web Credentials (2/2)
Input Capture (0/4)
Modify Authentication Process (1/9)
Multi-Factor Authentication Interception
Multi-Factor Authentication Request Generation
Network Sniffing
OS Credential Dumping (3/8)
Steal Application Access Token
Steal or Forge Authentication Certificates
Steal or Forge Kerberos Tickets (1/5)
Discovery 34 techniques
Account Discovery (2/4)
Application Window Discovery
Browser Information Discovery
Cloud Infrastructure Discovery
Cloud Service Dashboard
Cloud Service Discovery
Cloud Storage Object Discovery
Container and Resource Discovery
Debugger Evasion
Device Driver Discovery
Domain Trust Discovery
File and Directory Discovery
Group Policy Discovery
Local Storage Discovery
Log Enumeration
Network Service Discovery
Network Share Discovery
Network Sniffing
Password Policy Discovery
Peripheral Device Discovery
Permission Groups Discovery
Lateral Movement 9 techniques
Exploitation of Remote Services
Internal Spearphishing
Lateral Tool Transfer
Remote Service Session Hijacking (0/2)
Remote Services (4/8)
Replication Through Removable Media
Software Deployment Tools
Taint Shared Content
Use Alternate Authentication Material (3/4)
Collection 17 techniques
Adversary-in-the-Middle (0/4)
Archive Collected Data (1/3)
Audio Capture
Automated Collection
Browser Session Hijacking
Clipboard Data
Data from Cloud Storage
Data from Configuration Repository (0/2)
Data from Information Repositories (1/6)
Data from Local System
Data from Network Shared Drive
Data from Removable Media
Data Staged (1/2)
Email Collection (1/3)
Input Capture (0/4)
Screen Capture
Video Capture
Command and Control 18 techniques
Application Layer Protocol (1/5)
Communication Through Removable Media
Content Injection
Data Encoding (0/2)
Data Obfuscation (1/3)
Dynamic Resolution (0/3)
Encrypted Channel (0/2)
Fallback Channels
Hide Infrastructure
Ingress Tool Transfer
Multi-Stage Channels
Non-Application Layer Protocol
Non-Standard Port
Protocol Tunneling
Proxy (4/4)
Remote Access Tools (0/3)
Traffic Signaling (0/2)
Web Service (1/3)
Exfiltration 9 techniques
Automated Exfiltration (0/1)
Data Transfer Size Limits
Exfiltration Over Alternative Protocol (1/8)
Exfiltration Over C2 Channel
Exfiltration Over Other Network Medium (0/1)
Exfiltration Over Physical Medium (0/1)
Exfiltration Over Web Service (0/4)
Scheduled Transfer
Transfer Data to Cloud Account
Impact 15 techniques
Account Access Removal
Data Destruction (0/3)
Data Encrypted for Impact
Data Manipulation (0/3)
Defacement (0/2)
Disk Wipe (0/2)
Email Bombing
Endpoint Denial of Service (0/4)
Financial Theft
Firmware Corruption
Inhibit System Recovery
Network Denial of Service (0/2)
Resource Hijacking (0/4)
Service Stop
System Shutdown/Reboot
legend


FIN7
DarkHotel
GhostSec
Predatory Sparrow
Electrum
SideCopy
Selection Controls
Layer Controls
Technique Controls
Reconnaissance 11 techniques
Active Scanning (1/3)
Gather Victim Host Information (1/4)
Gather Victim Identity Information (2/3)
Gather Victim Network Information (1/6)
Gather Victim Org Information (1/4)
Phishing for Information (1/4)
Search Closed Sources (0/2)
Search Open Technical Databases (0/5)
Search Open Websites/Domains (0/3)
Search Threat Vendor Data
Search Victim-Owned Websites
Resource Development 8 techniques
Acquire Access
Acquire Infrastructure (2/8)
Compromise Accounts (1/3)
Compromise Infrastructure (2/6)
Develop Capabilities (1/4)
Establish Accounts (2/3)
Obtain Capabilities (2/7)
Stage Capabilities (1/6)
Initial Access 11 techniques
Content Injection
Drive-by Compromise
Exploit Public-Facing Application
External Remote Services
Hardware Additions
Phishing (2/4)
Replication Through Removable Media
Supply Chain Compromise (1/3)
Trusted Relationship
Valid Accounts (1/4)
Wi-Fi Networks
Execution 17 techniques
Cloud Administration Command
Command and Scripting Interpreter (3/13)
Container Administration Command
Deploy Container
ESXi Administration Command
Exploitation for Client Execution
Input Injection
Inter-Process Communication (0/3)
Native API
Poisoned Pipeline Execution
Scheduled Task/Job (1/5)
Serverless Execution
Shared Modules
Software Deployment Tools
System Services (0/3)
User Execution (2/3)
Windows Management
Persistence 23 techniques
Account Manipulation (4/7)
BITS Jobs
Boot or Logon Autostart Execution (0/14)
Boot or Logon Initialization Scripts (0/5)
Cloud Application Integration
Compromise Host Software Binary
Create Account (1/3)
Create or Modify System Process (2/5)
Event Triggered Execution (0/16)
Exclusive Control
External Remote Services
Hijack Execution Flow (0/12)
Implant Internal Image
Modify Authentication Process (0/9)
Modify Registry
Office Application Startup (0/6)
Power Settings
Privilege Escalation 14 techniques
Abuse Elevation Control Mechanism (0/6)
Access Token Manipulation (0/5)
Account Manipulation (0/5)
Boot or Logon Autostart Execution (0/14)
Boot or Logon Initialization Scripts (0/5)
Create or Modify System Process (2/5)
Domain or Tenant Policy Modification (1/2)
Escape to Host
Event Triggered Execution (0/16)
Exploitation for Privilege Escalation
Hijack Execution Flow (0/12)
Process Injection (0/12)
Scheduled Task/Job (1/5)
Valid Accounts (1/4)
Defense Evasion 47 techniques
Abuse Elevation Control Mechanism (0/6)
Access Token Manipulation (0/5)
BITS Jobs
Build Image on Host
Debugger Evasion
Delay Execution
Deobfuscate/Decode Files or Information
Deploy Container
Direct Volume Access
Domain or Tenant Policy Modification (1/2)
Email Spoofing
Execution Guardrails (0/2)
Exploitation for Defense Evasion
File and Directory Permissions Modification (0/2)
Hide Artifacts (0/14)
Hijack Execution Flow (0/12)
Impair Defenses (2/12)
Impersonation
Indicator Removal (1/10)
Indirect Command Execution
Masquerading
Credential Access 17 techniques
Adversary-in-the-Middle (0/4)
Brute Force (0/4)
Credentials from Password Stores (1/6)
Exploitation for Credential Access
Forced Authentication
Forge Web Credentials (0/2)
Input Capture (1/4)
Modify Authentication Process (0/9)
Multi-Factor Authentication Interception
Multi-Factor Authentication Request Generation
Network Sniffing
OS Credential Dumping (2/8)
Steal Application Access Token
Steal or Forge Authentication Certificates
Steal or Forge Kerberos Tickets (0/5)
Discovery 34 techniques
Account Discovery (2/4)
Application Window Discovery
Browser Information Discovery
Cloud Infrastructure Discovery
Cloud Service Dashboard
Cloud Service Discovery
Cloud Storage Object Discovery
Container and Resource Discovery
Debugger Evasion
Device Driver Discovery
Domain Trust Discovery
File and Directory Discovery
Group Policy Discovery
Local Storage Discovery
Log Enumeration
Network Service Discovery
Network Share Discovery
Network Sniffing
Password Policy Discovery
Peripheral Device Discovery
Permission Groups Discovery
Lateral Movement 9 techniques
Exploitation of Remote Services
Internal Spearphishing
Lateral Tool Transfer
Remote Service Session Hijacking (0/2)
Remote Services (1/8)
Replication Through Removable Media
Software Deployment Tools
Taint Shared Content
Use Alternate Authentication Material (0/4)
Collection 17 techniques
Adversary-in-the-Middle (0/4)
Archive Collected Data (0/3)
Audio Capture
Automated Collection
Browser Session Hijacking
Clipboard Data
Data from Cloud Storage
Data from Configuration Repository (0/2)
Data from Information Repositories (1/6)
Data from Local System
Data from Network Shared Drive
Data from Removable Media
Data Staged (0/2)
Email Collection (0/3)
Input Capture (1/4)
Screen Capture
Video Capture
Command and Control 18 techniques
Application Layer Protocol (1/5)
Communication Through Removable Media
Content Injection
Data Encoding (1/2)
Data Obfuscation (0/3)
Dynamic Resolution (0/3)
Encrypted Channel (0/2)
Fallback Channels
Hide Infrastructure
Ingress Tool Transfer
Multi-Stage Channels
Non-Application Layer Protocol
Non-Standard Port
Protocol Tunneling
Proxy (0/4)
Remote Access Tools (1/3)
Traffic Signaling (0/2)
Web Service (1/3)
Exfiltration 9 techniques
Automated Exfiltration (0/1)
Data Transfer Size Limits
Exfiltration Over Alternative Protocol (1/8)
Exfiltration Over C2 Channel
Exfiltration Over Other Network Medium (0/1)
Exfiltration Over Physical Medium (0/1)
Exfiltration Over Web Service (0/4)
Scheduled Transfer
Transfer Data to Cloud Account
Impact 15 techniques
Account Access Removal
Data Destruction (0/3)
Data Encrypted for Impact
Data Manipulation (0/3)
Defacement (1/2)
Disk Wipe (1/2)
Email Bombing
Endpoint Denial of Service (0/4)
Financial Theft
Firmware Corruption
Inhibit System Recovery
Network Denial of Service (0/2)
Resource Hijacking (0/4)
Service Stop
System Shutdown/Reboot
legend



KEY FINDINGS

Shared techniques include credential abuse, remote access, and web-based C2.

RECOMMENDATIONS

Monitor remote access, detect credential dumping, deploy EDR, segment IT/OT.